Drone-Based Farming

Drone-based farming uses unmanned aerial vehicles (UAVs) equipped with cameras, sensors, and sprayers to support different farm activities. Drones help in monitoring crop health, spraying fertilizers or pesticides, mapping fields, and analyzing soil conditions. However, drones are not commonly used for actual picking or harvesting of crops.

PROS	CONS
Reduces Labor Needs	High Initial Costs
Increases Efficiency	Requires Skillful Operators
Improves Crop Management	Weather Limitations
Saves Resources (Water, Chemicals)	Limited Battery Life
Fast Monitoring of Large Fields	Privacy and Regulation Issues

How Much Does Drone-Based Farming Cost, and What Smart Choices Can Farmers Make?

The cost of setting up drone-based farming depends on whether a farmer buys a drone or hires the service. Purchasing a basic spraying drone costs around ₹2–₹4 lakh, while advanced models can go up to ₹10 lakh. In addition, farmers need to spend around ₹1 lakh on setup, including operator training, mapping software, and trial runs. Annual maintenance can cost another ₹50,000–₹60,000.

Alternatively, many farmers choose to hire drone services, which cost ₹500–₹700 per acre per spraying session and ₹200–₹400 per acre for monitoring or mapping. For small and medium farmers, using service-based drones is often more practical and affordable, as it avoids heavy upfront costs and complex licensing.

Small farmers can also overcome the high cost of drones by coming together in cooperatives or farmer groups and sharing the equipment. By pooling their resources, they can afford advanced technology without each farmer needing to buy a drone individually. This shared approach makes drone use practical and cost-effective, allowing farmers to improve crop monitoring, reduce input costs, and increase yields, all while making modern tools accessible even to those with limited budgets.

How It's Used Abroad

- **Japan & China**: Used for precise spraying in rice and tea fields where manual access is difficult. Helps save a lot of labor in hilly areas.
- **USA**: Widely used for large-scale crop monitoring (corn, soybeans, wheat). Drones capture NDVI (Normalized Difference Vegetation Index) images to analyze plant health and predict yields.
- **Europe**: Used in vineyards and specialty crops for disease detection and targeted spraying.
- **Brazil:** Here soybean farmers use drones to cut pesticide use by about 30%, saving cash and helping the environment.

Why India Needs It Now?

- There aren't enough farm workers, and labor expenses are rising. Drones reduce the need for manual work and help manage large fields quickly.
- Climate change and erratic weather require quick, precise action to save crops.
- Common pest attacks require fast action. Drones can detect problems early through aerial images, helping farmers respond before big losses happen.
- Drones help optimize inputs and make farming more efficient and sustainable.
- New schemes and subsidies encourage precision farming and the use of drones, making it easier for farmers to adopt this technology now.

Market Size

The agriculture drone market is expanding rapidly in both India and globally, with India expected to outpace the global average in growth rate due to strong policy support, innovation, and a large agricultural base.

Region	Market Size (2024)	Projected Size (2032/33)	CAGR (approx.)
India	USD 243–245 million	USD 2.1 billion (2033)	24.1%
Global	USD 4.98 billion	USD 23.78 billion (2032)	18.5%–32%

Market Entry Strategies

Strategy	Description	Main Profit Source
Drone-as-a-Service (DaaS)	**Renting drones and providing spraying, mapping, or monitoring services to farmers, often on a per-acre or per-hour basis**	**Service fees for spraying, mapping, monitoring; recurring revenue from multiple clients**
B2B Partnerships	**Tie-ups with agri-input companies, farm equipment makers, or large farms to bundle drone services with other offerings**	**Contracted service fees, revenue-sharing, or bulk service agreements with partners**
Franchise/Entrepreneur	**Training and equipping local entrepreneurs or retailers to operate drones and offer services in rural areas**	**Franchise fees, equipment sales, ongoing royalties or a share of local service revenue**

Government Projects	**Participation in subsidy/demonstration programs and public-private partnerships, often for large-scale or pilot deployments**	**Government contracts, demonstration project funding, and grants/subsidies**
Data Analytics Services	**Bundling drone imagery with actionable insights for crop management, yield prediction, and advisory**	**Subscription fees, analytics/report sales, or premium advisory services for farmers or agri companies**

The most effective entry strategy is to combine DaaS with B2B partnerships and government engagement, focusing on affordable, scalable solutions and strong local presence. We can start with large-scale farms like tea gardens (Assam, West Bengal), sugarcane fields (Uttar Pradesh, Maharashtra), and banana plantations (Tamil Nadu) covering huge areas. Also, regions growing cash crops like cotton (Maharashtra, Telangana) and vegetables need frequent spraying to manage pests. Drones allow precise, safe, and efficient spraying, reducing chemical use and exposure risks.

Government Support

1. Subsidies under SMAM (Sub-Mission on Agricultural Mechanization)
Provides 40%–75% subsidy to help farmers and groups buy drones.

2. Drone Shakti Mission
Supports startups and service providers to offer drones as a service.

3. Kisan Drone Scheme
Promotes drone use among farmers for spraying and crop monitoring.

4. Several states, such as Maharashtra, Tamil Nadu, Punjab, Karnataka, Andhra Pradesh, and Telangana, offer additional subsidies, free training, or top-up incentives for specific groups or crops.

Is it all super easy?

No. It needs regulatory approvals, skilled operators, and trust-building. But, once the system is in place, scaling up is straightforward and demand-driven.

Disaster-related uses of agricultural drones

After floods, cyclones, or storms, drones can quickly fly over fields to assess crop damage, waterlogging, and soil erosion. This helps farmers and government agencies plan compensation, insurance payouts, and recovery faster.

These uses are most needed in flood-prone areas like Bihar, Assam, Eastern Uttar Pradesh, Odisha, and West Bengal; cyclone-affected coastal states like Andhra Pradesh, Tamil Nadu, Gujarat, and Odisha; drought-prone areas like Rajasthan, Gujarat, Maharashtra, and Karnataka; and hilly regions prone to landslides such as Uttarakhand, Himachal Pradesh, and Northeast states.

Top Agri-Drone Brands in India

Company	**Notable Products/Features**
Garuda Aerospace	Kisan Drone 16L/8L, DGCA-approved
ideaForge	Rugged, sensor-rich drones
Asteria Aerospace	AI-driven analytics, SkyDeck app
insideFPV	Krishi agri-drone, automation features

Paras Aerospace	Agricopter V2.1, data-driven solutions
Dhaksha	DH-AGRIGATOR-E10, customized UAVs
Drone Destination	Agristar 2.0, training and services
Marut Drones	RPTO-M, AG 365 S, multi-utility drones
IoTech	Agribot, AI-powered analytics
Thanos Technologies	Syena H10i, autonomous spraying